Venkata Lakshmi Parimala Pasupuleti

Forward Deployed Engineer · Agentic AI · LLMOps · GenAI Production Delivery · Python · AWS · Client-Embedded · 9yr Production

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PROFESSIONAL SUMMARY

Forward Deployed Engineer with 9+ years of production experience embedding with clients, translating high-value use cases into working GenAI and agentic AI solutions, and shipping from ambiguous requirements to production with speed and precision. At Magizhs, I owned the full FDE delivery lifecycle for an external client - gathered requirements directly from stakeholders, designed and built a production agentic AI system (LangGraph, FastAPI, Python) with RAG over regulatory documents, tool orchestration, and human-in-the-loop review gates, delivered to production in weeks as sole technical owner. LeaseI is the architectural proof point: an 8-agent LangGraph agentic platform with HITL safety gates, long-term memory, contextual reasoning, and deterministic guardrails - the same agentic workflow and human review pattern the Deloitte FDE practice deploys for enterprise clients. AgentProbe is the LLMOps layer: a production evaluation platform with 5-metric quality engine, CI-integrated regression gate, prompt management, and multi-provider LLM routing (Anthropic API, OpenAI API, Ollama) - reusable assets the FDE pod contributes back to the practice. At Entrata, I owned production enterprise systems on AWS (S3, EC2, RDS, Glue) for 9 years at 99.5% uptime with 100% PHI compliance and zero security incidents - the reliability and compliance standards enterprise clients require. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Authorized to work in the US; open to H-1B sponsorship.

TECHNICAL SKILLS

Agentic AI & GenAI Delivery: Python (9yr), LangGraph, LangChain, Multi-Agent Frameworks, Agent Orchestration, Agentic Workflows, HITL (Human-in-the-Loop), Guardrails, Prompt Engineering, Prompt Management, Anthropic API, OpenAI API, Ollama, MCP, Generative AI, LLMs, Claude Code

LLMOps & RAG: LLMOps, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Evaluation Frameworks, Model Monitoring, RAG Pipelines, Vector Databases, ChromaDB, Embeddings, Chunking Strategies, Semantic Search, NLP, BERT, spaCy, PyTorch, scikit-learn

Cloud & Enterprise Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, FastAPI, Microservices, REST APIs, Event-Driven Architecture, RabbitMQ, Redis, PostgreSQL, MySQL, Oracle, GitHub Actions CI/CD, PHI/HIPAA Compliance, Security, TypeScript, React.js, Linux, Git, Agile/Scrum

PROJECTS

Leasel: 8-Agent Agentic Platform - HITL, Long-Term Memory & Guardrails

Agentic AI · HITL · LLMOps · Production

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

End-to-end agentic AI platform - the architecture Deloitte's FDE practice deploys for enterprise workflow automation. **Agentic platform:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Human Escalation) with real conditional routing, dynamic task delegation by confidence score, and long-term SQLite knowledge repository shared across agents and sessions. **HITL guardrails:** deterministic escalation to human reviewer when confidence is below threshold; AI never auto-resolves regulated or compliance decisions without human approval - the safety boundary enterprise clients require. **LLMOps:** full audit logging, CI regression gate, 93.6% precision across 1,240 test cases; reusable reference implementation. Contributed back as open-source documentation and runbook.

AgentProbe: LLMOps Evaluation Platform - Reusable FDE Asset

LLMOps · Evaluation Frameworks · Model Monitoring · CI/CD

Python · FastAPI · LangGraph · ChromaDB · MCP · GitHub Actions CI/CD · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production LLMOps platform - reusable evaluation framework, prompt management, and model monitoring infrastructure that an FDE pod contributes back to the practice after each client engagement. **Evaluation framework:** 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency) with LLM-as-Judge and automated regression gate that blocks quality degradation before deployment. **Multi-provider LLM routing:** clean abstraction layer for Anthropic API, OpenAI API, and Ollama - swappable without prompt rewrites; transferable to Amazon Bedrock routing. **RAG infrastructure:** ChromaDB vector database, chunking strategy optimization, embedding selection. Zero regressions across 1,240+ test cases. Open-source, enterprise-deployable, fully documented with runbooks.

AI on FHIR: Enterprise System Integration - HIPAA-Compliant GenAI

Enterprise APIs · Cloud-Native · HIPAA · Security

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Cloud-native GenAI solution integrating LLMs with enterprise REST APIs at compliance-grade reliability - translates natural-language queries into structured FHIR API calls, enforces HIPAA-compliant RBAC with authentication and full data isolation, ~60% reduction in manual retrieval time. Demonstrates the enterprise system integration pattern FDE pods build for clients: consuming existing enterprise APIs and transforming them into AI-powered workflows without replacing the underlying systems.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Forward Deployed, Client-Embedded Production Delivery

- Owned the full FDE delivery lifecycle for an external client - scoped requirements directly with stakeholders, architected and built a production agentic AI system (LangGraph, FastAPI, Python) with RAG over regulatory documents, tool orchestration, validation guardrails, and human review gates; translated ambiguous business requirements into working enterprise automation delivered to production in weeks as sole technical owner. Contributed reusable code assets and runbook documentation back to the practice.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977); designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and software engineering; delivered complex computational research outputs to non-technical faculty stakeholders.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **AWS production infrastructure at enterprise scale** - AWS S3/EC2/RDS/Glue (production data pipelines); Docker microservices (50% scalability improvement, 30% deployment time reduction); RabbitMQ event-driven async architecture with retry logic and idempotency guarantees; 99.5% uptime across multi-tenant SaaS; zero security incidents across 7yr of production operation.
- **Enterprise data pipelines and compliance** - owned HUD federal compliance pipeline (180,000 domain records, eligibility and payment calculations for 100K+ housing units) for 7 years; 200+ SQL optimizations (30% performance gain); 100% PHI/HIPAA compliance with encrypted pipelines; built Python/Bash automation eliminating 80% of manual operational workload; full audit logging throughout.
- **Client-stakeholder delivery and mentoring** - 120+ Agile sprints with product, compliance, and operations stakeholders; 20+ full-stack production modules shipped end-to-end; code reviews (300+ peer reviews); internal debugging tools reducing triage time 40%; mentored 5+ engineers; leveraged Claude Code to accelerate engineering delivery.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017